3. Project Planning Phase

3.1 Introduction

The project planning phase establishes the objectives, scope, resources, activities, schedule, and risks of PocketSmart AI. Its purpose is to create a clear roadmap so that development activities are organized, resources are used effectively, and the project is completed within the specified time.

3.2 Project Objectives

- To develop an easy-to-use AI-powered budget and recommendation assistant.
- To provide secure user registration and authentication.
- To integrate Google Gemini for budget plans, insights, and recommendations.
- To enable income and expense tracking with smart categorization.
- To generate personalized savings plans and track goals.
- To maintain history and display financial progress through a dashboard.
- To help users spend wisely and build better saving habits.

3.3 Project Scope

The scope includes the design and development of a web-based budget and recommendation assistant. Users can create an account, log in, record income and expenses, generate AI budget plans, view spending insights, set savings goals, receive purchase recommendations, chat with the finance assistant, and view history and progress. Direct bank integration, payment processing, mobile apps, and voice input are outside the current scope and are listed as future enhancements.

3.4 Project Modules

- Authentication Module
- Expense and Income Tracking Module
- AI Budget Planner Module
- Spending Insights Module
- Savings Goal Module
- Smart Recommendation Module
- AI Finance Chat Module
- History and Dashboard Module

3.5 Resource Planning

Resources include the development computer, internet connection, Python and FastAPI, a web frontend, a relational database, a Google Gemini API key, Chart.js, Visual Studio Code, and Git/GitHub for version control.

3.6 Project Schedule

Phase	Activities	Duration
1	Brainstorming, problem identification, idea finalization	Week 1
2	Requirement analysis and documentation	Week 1–2
3	System design, architecture, database design	Week 2–3
4	Backend and frontend development, Gemini integration	Week 3–6
5	Testing and bug fixing	Week 6–7
6	Demonstration preparation and final documentation	Week 7–8

3.7 Risk Analysis

Risk	Impact	Mitigation
Gemini API downtime or rate limits	AI features temporarily unavailable	Error handling, retry logic, user-friendly messages
Inaccurate or unsuitable AI advice	User may make a poor financial decision	Clear prompts, disclaimer that output is not professional financial advice, encourage verification
Exposure of financial data	Privacy and security risk	Hashed passwords, JWT, per-user data access, HTTPS in deployment
API key exposure	Security and misuse risk	Store key in server environment variables
Schedule delays	Incomplete features	Prioritize core modules first
Slow responses	Poor user experience	Loading indicators, concise prompts